FinAgent-RAG: Adaptive Multi-Agent Retrieval-Augmented Generation for Financial Document Intelligence

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Table of Contents

Abstract

Financial organizations increasingly rely on AI systems to analyze annual reports, quarterly filings, and regulatory disclosures for financial analysis and decision-making. However, financial documents often contain lengthy narratives, numerical tables, and evidence distributed across multiple sections, making accurate retrieval and reasoning difficult for traditional question answering systems. Existing fixed-pipeline Retrieval-Augmented Generation approaches frequently struggle with ambiguous financial queries and multi-step reasoning tasks. The proposed study aims to develop and evaluate an adaptive multi-agent financial document question answering system using the FinanceBench dataset. The proposed framework will include document preprocessing, chunk generation, semantic retrieval, adaptive query routing, evidence selection, reasoning, answer generation, and verification stages. Multiple collaborative agents will dynamically adjust the retrieval and reasoning workflow according to query complexity to improve grounded financial answer generation.

Controlled benchmarking experiments will be conducted against baseline RAG pipelines under identical experimental conditions using metrics such as answer relevance, faithfulness, numerical correctness, reasoning consistency, and retrieval effectiveness. The proposed study is expected to demonstrate whether adaptive retrieval and multi-agent reasoning improve financial question answering performance for complex financial queries. The findings are intended to support the development of more reliable, explainable, and trustworthy AI systems for financial analysis and real-world financial decision-making tasks.

List of Figures

List of Tables

List of Abbreviations

- AI - Artificial Intelligence
- EM - Exact Match
- FQA – Financial Question Answering
- GPT - Generative Pre-trained Transformer
- HyDE - Hypothetical Document Embeddings
- LLM - Large Language Model
- MRR - Mean Reciprocal Rank
- PDF - Portable Document Format
- QA - Question Answering
- RAG - Retrieval-Augmented Generation
- SQL – Structured Query Language
- SEC - Securities and Exchange Commission
- Top-k - Top Ranked Retrieved Results
- Vector DB -Vector Database

1. Background

Financial reporting has traditionally been used by organisations to communicate their performance, risks, and financial position. Information related to revenue, expenditure, assets, liabilities, and future outlook is commonly presented through annual reports, quarterly filings, and earnings disclosures. Over the years, these documents have become larger and more detailed as reporting requirements and business operations have expanded. According (Jonker and Gomstyn, 2023), a significant portion of enterprise information is now stored in unstructured forms such as narrative reports and textual disclosures, which makes interpretation more difficult and time-consuming. Greater attention has therefore been given to methods that can assist in analysing financial documents more efficiently.

Some common characteristics of modern financial documents include:

- Financial information is often distributed across multiple sections rather than presented in one place.
- Numerical values are usually accompanied by long narrative explanations and disclosures.
- Important details may be embedded within textual descriptions, tables, or notes.
- Users often need to interpret related information together before reaching conclusions.

At the same time, major developments have been observed in language-based systems capable of processing large volumes of textual information. Earlier approaches were largely dependent on keyword matching and manually designed search methods. More recent studies have shown that financial language can now be interpreted with improved contextual understanding through domain-specific models trained on financial data (Mahendran et al., 2025). This shift has gradually changed how financial information is explored, particularly in situations where users need direct access to specific details hidden across long and complex reports.

Interest has also been directed towards approaches that connect generated responses with supporting evidence from source documents. Retrieval-based methods have been explored to improve answer quality by grounding outputs in external information rather than relying entirely on internal model knowledge (Wu et al., 2025b). Concerns related to evaluation and reliability have also been raised in recent studies, where generated responses, as discussed by (Malin et al., 2025), may appear convincing even when they are not fully supported by evidence, making assessment difficult in practice. In addition, research on language generation has reported that hallucination-related issues continue to affect the reliability of generated outputs, with unsupported responses still being observed in complex tasks(Ji et al., 2024).

Several broader challenges continue to be highlighted in current research:

- Retrieved information may not always directly support the generated response.
- Responses that appear fluent can still contain unsupported or incomplete details.
- Multi-step reasoning across financial documents remains difficult for many systems.

Questions that require reasoning across multiple parts of a document remain particularly challenging. These cases involve linking information, interpreting values, or combining details that are not located together. Research on reasoning systems shows that such tasks are still difficult, especially when information is distributed across sources (Wang et al., 2026). In addition, studies on language generation highlight that systems can produce responses that appear convincing even when they are not fully supported by available data, with hallucination rates reported to exceed 20% in some cases (Ji et al., 2024).

Survey-level research provides a broader understanding of how the field is developing. Reviews of financial text analysis, as explained by (Mahendran et al., 2025), show that much of the work has focused on sentiment analysis and prediction tasks, with less emphasis on document-based question answering.

Other studies that examine system behaviour and agent-based approaches point out challenges related to coordination, consistency, and reliability in complex tasks (Zhang et al., 2026). These observations suggest that, even with ongoing progress, understanding how systems perform on real financial documents remains an open area.

2. Problem Statement and Related Research

2.1 Problem Statement

Despite the growing availability of financial information, extracting reliable answers from financial documents remains difficult in practice. Annual reports and quarterly filings often contain interconnected information distributed across tables, disclosures, footnotes, and descriptive sections, where important numerical evidence may span multiple pages or cross-referenced statements. Structural issues such as cell shifting in tabular parsing, fragmented chunk boundaries, and semantic dilution of cross-reference footnotes may further reduce contextual continuity during retrieval.

As a result, users are frequently required to combine information from multiple sections before a clear financial interpretation can be formed. These challenges become more significant when queries require numerical comparison, multi-step reasoning, or interpretation across reporting periods.

Although Retrieval-Augmented Generation systems have shown improvements in document-based question answering, many existing approaches still rely on fixed retrieval pipelines that struggle to adapt to different query complexities and financial document structures. Most evaluations are conducted under controlled environments using well-structured queries, where retrieval accuracy is emphasized more than reasoning reliability, evidence grounding, contextual consistency, and numerical correctness. Consequently, generated responses may appear correct while important supporting evidence remains incomplete, weakly grounded, or semantically disconnected from the original document source. Existing evaluation approaches also do not consistently examine retrieval quality, reasoning consistency, and evidence support together under realistic financial question answering conditions.

The main problems identified in this research are:

- Financial evidence is frequently fragmented across tables, disclosures, footnotes, and cross-referenced report sections.
- Structural inconsistencies during document parsing and chunk generation may reduce retrieval precision and semantic continuity.
- Fixed retrieval pipelines struggle with multi-step financial reasoning and cross-section evidence aggregation.
- Existing evaluation methods do not consistently validate answer grounding, reasoning consistency, and numerical correctness together.
- Performance observed in controlled benchmark settings may not fully reflect behaviour on realistic financial document question answering tasks.

The study therefore focuses on developing an adaptive multi-agent retrieval and reasoning framework capable of handling complex financial document structures and fragmented evidence distributions. Specialized agents will be responsible for query refinement, contextual retrieval, evidence filtering, reasoning, and verification in order to address schema-level inconsistencies, cross-referenced disclosures, and multi-section financial reasoning challenges more effectively.

2.2 Literature Review

The literature review presents studies related to retrieval-based systems, agent-driven frameworks, and financial document question answering. The reviewed research shows the progression from traditional retrieval pipelines to more adaptive reasoning approaches designed to improve retrieval quality and answer generation. Several limitations were still identified in areas such as multi-step reasoning, dynamic query handling, and evidence grounding within financial documents. These gaps highlight the need for a more adaptive and reliable financial question answering framework.

(Choi et al., 2025) introduced FinAgentBench for evaluating agentic retrieval in financial question answering. Around 26,000 expert-annotated samples from S&P 500 filings were used to test multi-step retrieval reasoning. The study separated document-level and chunk-level retrieval, which helped analyse retrieval behaviour more clearly in financial environments. However, grounded answer generation and hallucination control were not explored deeply. This limitation is relevant because the present study evaluates both retrieval quality and evidence-supported financial answer generation together.

(Govind Srinivasan et al., 2025) proposed a financial RAG framework using agentic AI and Multi-HyDE retrieval. Multiple hypothetical queries were generated to improve retrieval coverage and reduce hallucinations in financial question answering tasks. Hybrid retrieval methods combining dense and sparse retrieval were also integrated into the system. Although answer accuracy improved, the framework introduced higher workflow complexity and processing overhead. This gap is important because adaptive routing and controlled retrieval benchmarking are still required for handling different levels of financial query complexity.

(Alam et al., 2025) developed AstuteRAG-FQA, a task-aware framework for financial question answering using hybrid retrieval and adaptive prompting. Financial queries were categorized into multiple reasoning levels ranging from direct factual retrieval to causal reasoning tasks. The study highlighted the importance of adaptive retrieval behaviour in regulated financial environments. However, stronger focus was placed on compliance and proprietary settings rather than retrieval grounding and reasoning evaluation together. This limitation is relevant because the present study jointly analyses retrieval quality, reasoning consistency, and hallucination behaviour.

(Cheng et al., 2026) proposed a retrieval-augmented system for answering questions from financial 10-K reports. Hybrid retrieval and neural reranking were used to improve evidence selection from lengthy financial documents. The study showed that reranking significantly improved answer correctness and reduced retrieval errors. Even though retrieval refinement was improved, adaptive reasoning workflows and query complexity handling were not analysed deeply. This gap supports the present study because complex financial questions often require dynamic retrieval and reasoning behaviour.

(Harsha et al., 2026) presented a multi-agent pipeline for generating synthetic financial question-answer datasets from business documents. Separate agents were used for extraction, selection, question generation, answer generation, and validation tasks. The framework reduced dependence on manually created financial QA datasets and improved scalability of data generation. However, the work mainly focused on dataset creation and fine-tuning rather than adaptive retrieval and grounded response generation during inference. This limitation is relevant because the present study focuses on retrieval reasoning and evidence-supported financial question answering workflows.

(Kartheesan et al., 2026) proposed a cloud-based personal finance assistant using Retrieval-Augmented Generation and SQL agents. The framework combined database querying with retrieval grounding to reduce hallucinations in financial conversations. Budget tracking, expense analysis, and personalized financial recommendations were also supported. However, the study mainly focused on reliable database interaction rather than adaptive reasoning workflows. This limitation is relevant because the present study focuses on adaptive financial retrieval and reasoning together.

(Xu et al., 2025) introduced Agentic RAG with Human-in-the-Retrieval, where domain experts were treated as active retrieval sources alongside knowledge bases. The framework highlighted how human expertise could improve factual grounding and contextual understanding in complex domains. Autonomous orchestration workflows using retrieval and generation agents were also discussed. However, the study remained mostly conceptual and lacked detailed evaluation in financial reasoning environments. This gap is relevant because the present study focuses on practical and benchmark-driven financial question answering systems.

(Zhang et al., 2026) proposed ACE, an Agentic Context Engineering framework for self-improving language models. The framework used generation, reflection, and curation modules to preserve detailed contextual knowledge while reducing context collapse. Strong improvements were reported across finance and reasoning benchmarks with lower adaptation cost. However, the study concentrated mainly on context evolution and prompt optimization instead of retrieval-grounded financial question answering. This limitation is important because financial systems require both adaptive context handling and reliable retrieval workflows.

(Kukreja et al., 2025) proposed a relevance-aware Agentic RAG framework using Relevance Generative Answering (RGA). A relevance detection block was introduced to better understand user intent and improve contextual response generation. Improvements in precision, recall, and response relevance were reported when compared with traditional Agentic RAG systems. However, the framework mainly emphasized intent handling and response refinement rather than grounded financial reasoning and explainability. This gap is relevant because the present study evaluates both retrieval reliability and financial reasoning consistency.

(Wu et al., 2025a) proposed MultiRAG, a knowledge-guided framework designed to reduce hallucinations in multi-source Retrieval-Augmented Generation systems. Multi-source line graphs and confidence-based retrieval mechanisms were introduced to improve retrieval reliability across structured and unstructured data. Experimental results showed better performance in multi-hop and multi-source retrieval tasks. However, the work mainly focused on retrieval consistency and hallucination mitigation rather than adaptive financial reasoning workflows. This limitation is relevant because the present study combines retrieval grounding with context-aware financial reasoning.

(Shinde and Khanvilkar, 2025) proposed a secure Graph-RAG framework for financial compliance question answering using knowledge graphs, access control, and automated redaction. The system linked regulations, internal policies, and compliance records to support secure retrieval and privacy-aware reasoning. Experimental results showed strong retrieval accuracy and redaction performance in regulated financial environments. However, the framework mainly emphasized compliance security and controlled retrieval rather than adaptive reasoning or dynamic financial query orchestration. This limitation is relevant because the present study investigates both reliable reasoning and context-aware financial response generation.

(Hashmi, 2026) proposed Adaptive Hybrid Retrieval, a tier-based query routing framework combining Vector RAG, Tree Reasoning, and hybrid retrieval strategies across financial, legal, and medical documents. The framework dynamically selected retrieval mechanisms based on query complexity and demonstrated strong performance on FinanceBench and cross-reference reasoning tasks. Although retrieval adaptability improved significantly, the study mainly focused on retrieval optimization rather than financial reasoning consistency or personalized response generation. This gap is important because financial AI systems require both accurate retrieval and domain-aware reasoning capabilities.

(Ur Rahman et al., 2025) introduced MASQRAD, a multiagent actor-critic generative AI framework for resolving ambiguous user queries and generating actionable insights. The architecture used Actor, Critic, and Expert Analysis agents to refine queries, validate outputs, and generate analytical interpretations with reduced hallucination risk. The framework achieved strong performance in natural language visualization and data interpretation tasks. However, the work primarily focused on analytical query resolution and visualization generation rather than retrieval-grounded financial question answering workflows. This limitation is relevant because the current study emphasizes reliable retrieval orchestration and evidence-grounded financial reasoning.

(Hua et al., 2025) proposed a RAG-enhanced AI agent chatbot architecture for insurance operations using lightweight language models and vector retrieval mechanisms. The framework improved semantic understanding, contextual integrity, and insurance-specific reasoning through integration of retrieval and generation modules. Experimental evaluation demonstrated better response quality compared with traditional rule-based systems. Nevertheless, the architecture mainly focused on insurance customer support scenarios and did not deeply address multi-hop reasoning or adaptive retrieval orchestration. This gap is significant because complex financial question answering systems require stronger reasoning and retrieval coordination capabilities.

(Xiao et al., 2025) proposed the Reliable Reasoning Path framework to improve LLM reasoning using structured knowledge graphs and refined reasoning paths. The framework generated semantically and structurally coherent reasoning chains while filtering redundant or conflicting paths through a rethinking module. Experimental results showed improved reasoning performance and reduced hallucinations on knowledge-intensive tasks. However, the work concentrated primarily on reasoning path generation and knowledge graph integration rather than domain-specific financial retrieval optimization.

(Rao et al., 2025) proposed ContractIQ, a multimodal RAG-based agentic system for intelligent legal contract understanding using hybrid retrieval, Chain-of-Thought reasoning, and multi-agent coordination. The framework supported clause extraction, compliance analysis, summarization, and legal question answering with improved contextual interpretation.. However, the study primarily targeted legal contract analysis and did not evaluate scalability for dynamic financial reasoning or real-time financial knowledge integration. This limitation is important because financial AI systems require adaptive retrieval and continuously evolving financial knowledge processing.

Table 2.1 Comparative Evaluation Matrix of Existing Financial Retrieval and Reasoning Frameworks

Study	Retrieval Type	Context Window Handling	Tabular Parsing Support	Evaluation Dataset	Key Limitation
(Choi et al., 2025)	Dense / Agentic Retrieval	Two-stage document and chunk-level ranking	Not Explicitly Addressed	Fin Agent Bench (26K expert-annotated samples)	Focuses mainly on retrieval benchmarking rather than grounded answer generation and verification
(Govind Srinivasan et al., 2025)	Hybrid Retrieval (Dense + BM25 + Multi-HyDE)	Multi-step retrieval with agentic orchestration	Limited table-aware retrieval support	Financial QA Benchmarks / FinanceBench	Increased architectural complexity and higher retrieval orchestration overhead
(Alam et al., 2025)	Hybrid Retrieval	Task-aware adaptive prompt routing	Partial support for financial and proprietary structured data	Proprietary + Public Financial Data	Strong compliance focus but limited retrieval benchmarking under identical preprocessing conditions
(Cheng et al., 2026)	Hybrid Retrieval with Neural Reranking	Fixed-size chunking with overlap strategy	Limited support for financial tables	FinDER Benchmark Dataset	Focuses primarily on reranking effectiveness rather than adaptive reasoning workflows
(Harsha et al., 2026)	Agentic Retrieval and Content Selection	Multi-agent content extraction and selection workflow	Strong multimodal text-table extraction support	FinQA, TAT-QA, FinTabNet	Focuses on synthetic QA generation rather than real-time financial question answering
(Kartheesan et al., 2026)	Hybrid Retrieval with SQL-Agent Integration	Context grounding through SQL + RAG orchestration	Strong support for structured relational financial data	Conceptual architecture / No standardized benchmark	Lacks empirical benchmarking and real-world deployment validation
(Xu et al., 2025)	Agentic Retrieval with Human-in-the-Loop	Dynamic iterative retrieval orchestration	Not specifically focused on tabular financial parsing	Conceptual agentic RAG scenarios	Human dependency increases operational complexity and scalability concerns
(Zhang et al., 2026)	Context-adaptive Agentic Retrieval	Evolving long-context playbook strategy with incremental updates	Limited direct financial table reasoning support	AppWorld, FiNER, Numerical Reasoning Benchmarks	High context growth may increase inference and memory overhead
(Kukreja et al., 2025)	Agentic Retrieval with Relevance-based Filtering	Intent-aware iterative query refinement	Limited structured financial table integration	Experimental relevance evaluation datasets	Focuses mainly on relevance optimization rather than evidence traceability
(Wu et al., 2025a)	Multi-source Hybrid Retrieval with Knowledge Graphs	Multi-source graph-based context aggregation	Supports structured, semi & unstructured data	Multi-domain retrieval + Multi-hop QA datasets	Increased graph construction complexity and computational overhead

(Shinde and Khanvilkar, 2025)	Graph-based Hybrid Retrieval	Structured graph-linked retrieval with audit-aware filtering	Moderate support through structured compliance linking	FinQA, ObliQA, 10-K filings, synthetic KYC/AML datasets	Focuses heavily on security and compliance rather than adaptive reasoning
(Hashmi, 2026)	Adaptive Hybrid Retrieval	Tier-based dynamic routing for varying query complexity	Strong support for hierarchical document structures	FinanceBench, SEC filings, legal & medical datasets	Increased routing and orchestration complexity may affect scalability
(Ur Rahman et al., 2025)	Multi-agent Retrieval and Query Refinement	Actor-Critic iterative reasoning workflow	Limited direct financial table parsing support	Large-scale visualization and query datasets	Primarily focused on query refinement and visualization rather than document QA
(Hua et al., 2025)	Dense Vector-based RAG	Chunk-based contextual retrieval with overlap management	Moderate support for structured insurance knowledge	Insurance policy and operations datasets	Domain-specific implementation limits broader financial generalization
(Xiao et al., 2025)	Knowledge Graph-guided Retrieval	Multi-hop reasoning path generation & refinement	Limited support for explicit financial tabular parsing	Public reasoning and KG benchmark datasets	Graph reasoning introduces higher computational complexity
(Rao et al., 2025)	Hybrid Retrieval using BM25 + Dense Embeddings	Multi-agent Chain-of-Thought legal reasoning	Moderate support through legal clause extraction	CUAD benchmark dataset	Primarily optimized for legal contracts rather than financial analytical reasoning

Comparative Synthesis and Discussion

Existing studies in Retrieval-Augmented Generation and financial question answering mainly focus on isolated improvements such as hybrid retrieval optimization, graph-based reasoning, adaptive query routing, or hallucination reduction. Several frameworks improve semantic retrieval accuracy through dense or hybrid retrieval methods, while others emphasize multi-agent orchestration and reasoning enhancement. Some studies also introduce compliance-aware retrieval and evidence filtering to improve answer reliability in regulated domains. However, these capabilities are often evaluated separately rather than within a unified financial reasoning framework.

Context management strategies also vary considerably across existing systems. Some studies rely on fixed chunking methods, whereas others adopt hierarchical traversal, graph reasoning, or adaptive routing approaches for long-document retrieval. The comparative analysis further shows that explicit support for financial document structures remains limited in many existing frameworks. Although several studies handle structured or semi-structured data, challenges related to tabular parsing, fragmented numerical evidence, and cross-referenced disclosures are not consistently addressed.

Multimodal retrieval, evidence grounding, and verification-based reasoning are also integrated unevenly across current financial question answering systems. As a result, retrieval quality and reasoning consistency may still degrade when handling complex multi-section financial queries.

Key limitations identified across the reviewed studies include:

- Inconsistent preprocessing, chunking, and evaluation strategies across different experiments.
- Limited benchmarking of retrieval quality, reasoning consistency, and hallucination control together.
- Weak integration of adaptive routing, evidence verification, and financial grounding within a single framework.

Another important observation is that most existing frameworks are evaluated using different datasets, retrieval architectures, and experimental settings, making direct comparison difficult. Many reported improvements therefore remain dependent on specific preprocessing pipelines or benchmark configurations. Limited research has systematically evaluated adaptive retrieval, multi-agent reasoning, evidence verification, and financial grounding together under identical preprocessing and evaluation conditions using realistic financial datasets such as FinanceBench. The proposed study aims to address this gap through a controlled benchmarking framework that jointly evaluates retrieval effectiveness, reasoning quality, numerical correctness, hallucination behaviour, and grounded financial answer generation.

2.3 Research Gaps and Critical Analysis

Research Gaps

Based on the reviewed literature, the following research gaps have been identified:

- Existing financial question answering studies often focus on answer accuracy, while giving less attention to whether answers are fully supported by precise evidence from financial documents such as reports and filings.
- Many studies explore retrieval methods or reasoning techniques independently, but fewer works compare multiple approaches under the same financial dataset, pre-processing setup, and evaluation metrics.
- Retrieval quality and answer generation are frequently evaluated separately, even though errors in retrieval directly affect reasoning accuracy and final answer correctness in financial QA.
- Explainability is often limited to displaying retrieved documents or reasoning traces, which does not guarantee that answers are grounded in the correct financial evidence or numerical values.
- Hallucination evaluation remains limited in financial QA, where different types of errors, such as incorrect numerical values or misinterpretation of financial statements, have varying levels of impact.

Critical Analysis

The reviewed studies demonstrated a gradual transition from fixed retrieval pipelines toward more adaptive retrieval and reasoning architectures. Recent approaches introduced hybrid retrieval, query rewriting, graph reasoning, adaptive routing, and multi-agent orchestration to improve financial question answering performance. However, most frameworks were evaluated using different preprocessing methods, chunking strategies, retrieval settings, and datasets, making direct comparison difficult. Many reported improvements therefore remained highly dependent on specific experimental configurations rather than standardized benchmarking conditions.

The comparative analysis also showed that existing studies mainly focused on isolated improvements such as retrieval optimization, hallucination reduction, compliance filtering, or reasoning enhancement rather than fully integrated financial reasoning frameworks. Explicit integration of adaptive retrieval, tabular financial parsing, evidence verification, reasoning consistency, and grounded answer generation within a single controlled framework remained limited across the reviewed literature.

Key research gaps identified across the reviewed studies include:

- Limited handling of fragmented numerical evidence distributed across tables, disclosures, and cross-referenced sections.
- Inconsistent evaluation of retrieval quality, hallucination behaviour, reasoning consistency, and numerical correctness together.
- Weak benchmarking consistency caused by variations in preprocessing pipelines and retrieval configurations.
- Limited use of verification-based workflows for validating evidence grounding before answer generation.

The literature therefore indicates the need for a more adaptive and systematically benchmarked financial question answering framework capable of handling complex financial document structures under identical preprocessing and evaluation conditions. The proposed study aims to address these limitations through an adaptive multi-agent Retrieval-Augmented Generation framework that combines query refinement, contextual retrieval, evidence filtering, reasoning, and verification within a unified benchmarking environment.

3. Research Questions

- How effectively will the proposed adaptive multi-agent framework retrieve relevant financial evidence and generate grounded answers from the FinanceBench dataset under controlled conditions?
- Which retrieval approach among Direct Language Model, Naïve Retrieval Pipeline, Query-Rewritten Retrieval, Multi-Query Retrieval & Adaptive Multi-Agent Retrieval will produce the most accurate, faithful, and context-supported financial answers when tested using the same dataset, embedding model, language model, and evaluation metrics?

- How will query complexity affect retrieval quality, numerical reasoning, and evidence grounding in financial document question answering tasks?
- What impact will adaptive query routing, query refinement, and multi-agent reasoning have on hallucination reduction and evidence-supported answer generation in financial documents?

4. Aims and Objectives

Aim

The aim of this research is to develop and benchmark an adaptive multi-agent retrieval and reasoning framework for financial document question answering using the FinanceBench dataset. The proposed system will use specialized agents for query understanding, retrieval, reasoning, and verification to generate evidence-grounded financial answers. The study will also examine whether adaptive query routing and multi-agent reasoning improve answer quality, retrieval effectiveness, and hallucination control when compared with traditional fixed retrieval pipelines.

Objectives

- To preprocess and prepare the FinanceBench dataset by extracting financial report text, creating document chunks, and attaching metadata such as company name, year, section, and page number.
- To implement Direct Language Model, Naïve Retrieval Pipeline, Query-Rewritten Retrieval, Multi-Query Retrieval, and Adaptive Multi-Agent Retrieval systems under the same experimental setup for fair comparison.
- To design an adaptive query routing mechanism that classifies financial questions into simple, moderate, and complex categories based on retrieval and reasoning requirements.
- To evaluate and statistically compare all systems using metrics such as answer relevance, faithfulness, context recall, numerical accuracy, hallucination rate, and response latency in order to measure whether the proposed adaptive multi-agent framework improves retrieval quality and reduces hallucinated financial responses compared with baseline RAG models.
- To compare the effectiveness of adaptive multi-agent retrieval and reasoning against fixed retrieval pipelines in handling financial document question answering tasks involving numerical extraction, comparison, and multi-section reasoning.

5. Significance of the Study

Financial reports contain large amounts of business information, yet extracting precise answers from these documents remains difficult. Important evidence is often distributed across tables, disclosures, footnotes, and multiple report sections, making complex financial questions challenging for traditional retrieval systems. Fixed retrieval pipelines may therefore return partially relevant evidence, leading to responses that appear convincing while important financial context is still missing.

The study is expected to provide a clearer understanding of how adaptive retrieval workflows improve financial question answering under realistic retrieval and reasoning conditions. Multiple retrieval pipelines will be benchmarked under identical experimental settings using shared datasets, embedding models, vector databases, and evaluation metrics. Retrieval quality, reasoning consistency, hallucination behaviour, numerical correctness, and evidence grounding will be analysed together rather than independently.

From a practical perspective, the proposed framework may support FinTech developers and organizations building trustworthy financial AI systems through:

- Improved evidence traceability and citation correctness during financial answer generation.
- A verification agent capable of validating retrieved evidence before final responses are generated to reduce unsupported claims and incorrect numerical interpretation.
- Stronger grounded retrieval and reasoning for evidence-sensitive financial analysis tasks.

The proposed verification and evidence-grounding workflow may also support future adaptation to high-risk compliance environments such as legal and healthcare document systems. In these domains, incorrect citation mapping, unsupported explanations, and missing evidence can directly affect regulatory compliance, operational reliability, and decision-making accuracy. The study therefore contributes not only to financial document question answering, but also to the broader development of trustworthy retrieval and reasoning systems for evidence-sensitive applications.

6. Scope of the Study

In this section main focus is on developing and evaluating an adaptive multi-agent retrieval and reasoning framework for financial document question answering using the FinanceBench dataset. The scope has been limited to controlled benchmarking of retrieval and reasoning workflows so that system performance can be analysed clearly under the same experimental conditions. Both the covered areas and excluded areas of the research are presented below to define the boundaries of the proposed work.

6.1 In-Scope of the Study

- The proposed framework will focus on financial document question answering using annual reports, financial filings, and supporting evidence from the FinanceBench dataset.
- The proposed framework will include:
    - query understanding
    - query refinement
    - adaptive routing
    - retrieval & evidence filtering
    - reasoning
    - answer verification agents

- Multiple retrieval approaches will be compared under the same experimental setup to analyse retrieval quality, reasoning performance, and answer grounding.
- Financial questions involving numerical extraction, comparison, trend analysis, and multi-section reasoning will be included within the evaluation scope.
- Evaluation will cover retrieval quality, answer relevance, faithfulness, context recall, hallucination rate, numerical accuracy, and reasoning consistency.
- The study will analyse how adaptive routing changes retrieval and reasoning behaviour for simple, moderate, and complex financial queries.

6.2 Out-of-Scope of the Study

- The study will not involve real-world deployment or testing with live financial customers.
- Proprietary enterprise financial datasets will not be used since the research is limited to publicly available benchmark data.
- The work will focus only on text-based financial documents and will not analyse graphical financial charts or scanned handwritten reports.
- Advanced areas such as autonomous trading systems, portfolio optimization, or financial forecasting are outside the boundaries of the proposed research.

7. Research Methodology

7.1 Research Introduction

A structured experimental methodology will be followed to design and evaluate an adaptive multi-agent Retrieval-Augmented Generation system for financial document question answering. The study will focus on improving retrieval quality, reasoning capability, and answer grounding when processing annual reports, quarterly filings, and regulatory disclosures. The FinanceBench dataset will be used because it contains real financial questions, supporting evidence, and publicly available company filings. The dataset also represents realistic financial analysis scenarios where relevant information is distributed across multiple sections of a document.

The methodology will include financial document preprocessing, embedding generation, semantic retrieval, and adaptive multi-agent reasoning stages. Incoming queries will be analyzed according to their complexity level before retrieval and reasoning operations are performed. The generated responses will finally be benchmarked against baseline RAG pipelines using metrics such as answer relevance, faithfulness, context recall, reasoning consistency, and answer correctness.

The overall methodology flow of the proposed study is summarized below:

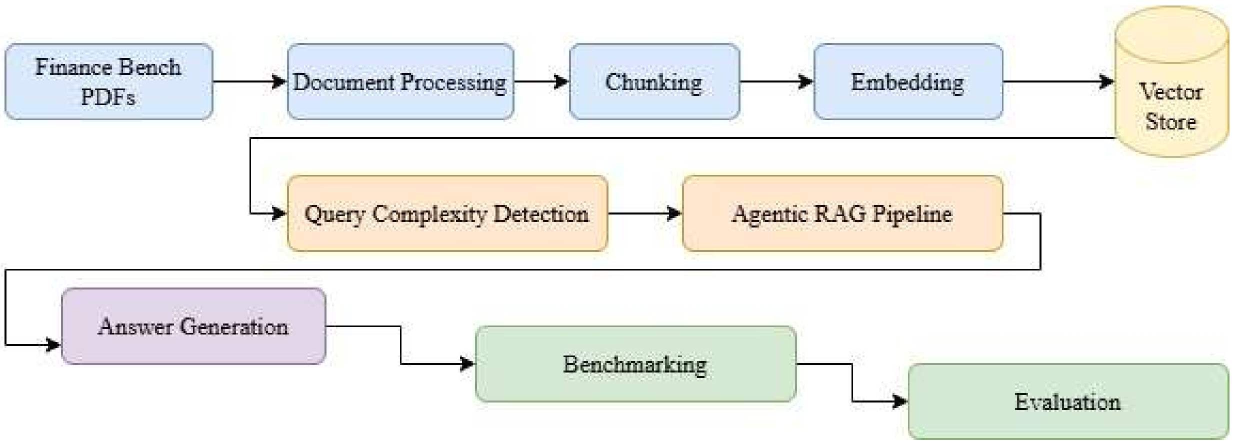


Figure 7.1.1 Research Methodology Overall Flow

7.2 Research Approach

A positivist research philosophy will mainly be followed in the study because the proposed financial question answering system will be evaluated using measurable and objective performance metrics. The research will focus on designing and benchmarking an adaptive multi-agent RAG framework under controlled experimental conditions. Metrics such as answer relevance, faithfulness, context recall, semantic similarity, exact match, and response latency will be used to evaluate system performance. Comparative analysis will also be performed against baseline RAG pipelines to measure retrieval and reasoning effectiveness.

The study will also include a limited interpretive component because financial question answering often involves complex reasoning over financial statements, disclosures, and numerical evidence. In certain cases, automated evaluation metrics alone may not fully capture reasoning validity or evidence grounding quality. Selected responses may therefore be manually examined to verify contextual correctness, reasoning consistency, and alignment with retrieved financial evidence. This combined approach will support both quantitative evaluation and qualitative validation of generated financial answers.

Table 7.1 Research Philosophy and Methodological Position

Main Philosophy	**Positivism**
Reason	The proposed system will be benchmarked using measurable evaluation metrics and controlled experimental analysis.
Supporting View	Interpretive Review
Reason	Financial reasoning quality and evidence grounding may require limited manual verification and contextual assessment.
Research Nature	Experimental and System-Development Based
Research Focus	Adaptive Multi-Agent RAG for Financial Document Question Answering
Evaluation Style	Quantitative benchmarking with selective qualitative review
Expected Output	A benchmarked and evaluated multi-agent RAG framework for financial QA

7.3 Research Strategy

The study will adopt a comparative experimental research design to develop and evaluate an adaptive multi-agent RAG system for financial document question answering. The proposed framework will be compared with baseline RAG pipelines to determine whether adaptive retrieval and multi-agent reasoning improve overall system performance. Structured benchmarking and controlled evaluation will also be carried out across multiple RAG configurations.

The experimental design will be important because retrieval quality, reasoning capability, and evidence grounding are closely interconnected in financial QA tasks. Controlled comparison between adaptive and non-adaptive RAG systems will therefore help evaluate the effectiveness of dynamic retrieval and multi-agent reasoning strategies. Performance evaluation will focus on answer relevance, faithfulness, context recall, retrieval accuracy, numerical correctness, semantic similarity, and response latency.

A controlled comparison methodology will be followed to maintain experimental consistency and improve the reliability of comparative findings. Components such as the FinanceBench dataset, financial documents, embedding generation, chunking strategy, vector storage mechanism, and base language model will remain fixed across experiments. Only the retrieval and reasoning pipeline design will be varied between different RAG configurations during evaluation. Overall, the research strategy will combine system development, benchmarking, experimentation, and performance analysis to evaluate improvements in grounded financial question answering.

Table 7.2 Controlled Experimental Design Overview

Experimental Element	Description
Research Design	Comparative Experimental Research
Primary Objective	Compare adaptive multi-agent RAG with baseline RAG systems
Independent Variable	RAG pipeline design and retrieval strategy
Dependent Variables	Answer relevance, faithfulness, context recall, retrieval accuracy, numerical correctness, semantic similarity, and response latency
Dataset Used	FinanceBench
Controlled Components	Dataset, document collection, chunking strategy, embedding model, vector database, language model, and evaluation questions
Comparison Style	Controlled benchmarking under identical experimental settings

7.4 Dataset Details

7.4.1 Dataset Description & Structure

The present study uses the Finance Bench dataset, which is designed for financial document question answering and retrieval-based evaluation. The dataset contains financial questions, reference answers, and supporting evidence collected from real company filings and reports.

Unlike general-purpose question answering datasets, Finance Bench focuses specifically on financial analysis scenarios. The questions are derived from realistic financial tasks that involve interpreting annual reports, quarterly filings, disclosures, and financial statements.

https://github.com/patronus-ai/financebench/tree/main/pdfs

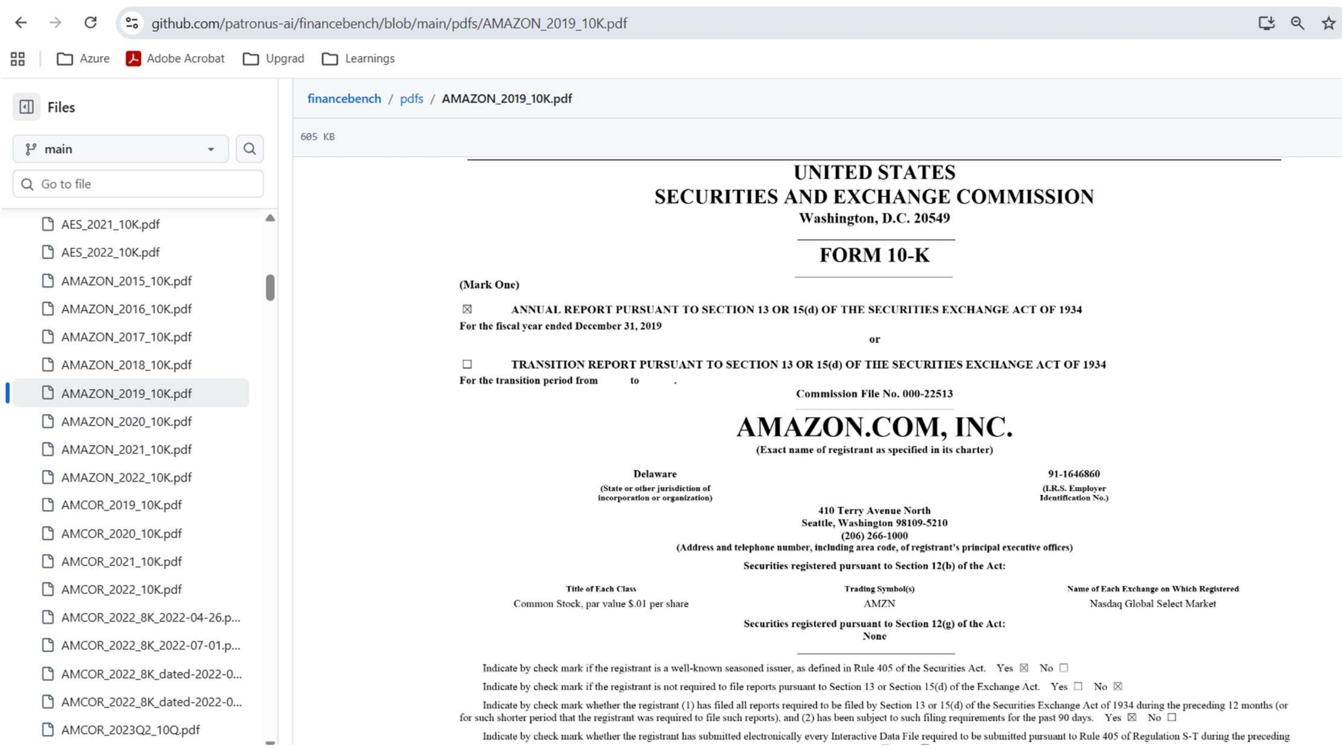


Figure 7.4.1 Finance Bench Sample Data

Finance Bench is constructed using real-world financial filings and associated question-answer pairs. Each dataset entry connects a financial question with its corresponding answer and supporting evidence extracted from a source document.

The source documents include multiple filing types such as:

- 10-K annual reports
- 10-Q quarterly reports
- 8-K filings
- Earnings reports
- Regulatory disclosures

The dataset combines both structured and unstructured financial information. Structured content includes balance sheets, income statements, and tabular financial records, while unstructured content includes management discussions, disclosures, and narrative explanations. Since relevant evidence may appear across different pages and sections, the dataset supports evaluation of retrieval systems at document, page, and chunk levels.

Another important aspect of the dataset is the inclusion of evidence annotations. These annotations identify the relevant portion of the financial document that supports the correct answer, enabling detailed evaluation of evidence grounding and retrieval effectiveness.

7.4.2 Feature-wise Description

The important dataset features used in this study are described below.

Table 7.3 Financial Bench Dataset Features Description

Feature Name	Description
question	Contains the financial question asked by the user. Questions may involve factual retrieval, numerical reasoning, comparison, or interpretation of financial information.
answer	Represents the ground-truth answer associated with the question. Answers may be textual, numerical, or explanatory depending on the query type.
evidence	Stores the supporting evidence extracted from the source financial document. This field is important for evaluating evidence grounding and answer faithfulness.
page_number	Identifies the page where the supporting evidence appears in the financial filing. This supports fine-grained retrieval evaluation.
company	Specifies the company associated with the financial document and question-answer pair.
document_type	Indicates the type of source document, such as 10-K, 10-Q, 8-K, or earnings report.
document_name	Contains the identifier or filename of the financial report used in the dataset.
document_year	Represents the year associated with the financial filing.
question_type	Categorizes questions based on complexity or reasoning requirements.
gics_sector	Indicates the industry sector classification of the company.
justification	Provides explanation or reasoning information for certain question-answer pairs.

7.5 Financial Document Processing and Knowledge Base Preparation

Financial documents used in the study will be collected from the FinanceBench dataset and processed before retrieval and indexing. Since financial reports contain both structured and unstructured information, a document preparation pipeline will be required to handle narrative discussions, financial tables, disclosures, footnotes, and section-wise analysis distributed across lengthy reports. The preprocessing workflow will begin with PDF text extraction while preserving important structural elements such as headings, tables, and section boundaries. Financial tables will receive particular attention because many financial questions require accurate numerical retrieval from statements such as balance sheets, income statements, and cash flow reports.

After extraction, the documents will undergo cleaning to remove repeated headers, page numbers, navigation elements, and other irrelevant content that may reduce retrieval quality. Important report sections such as management discussions, financial statements, notes to accounts, and risk disclosures will then be identified before the documents are divided into smaller retrieval-ready chunks.

Each chunk will be associated with metadata including company name, filing year, report type, page number, and section name to improve traceability and evidence grounding. Finally, the processed chunks and metadata will be stored within the retrieval knowledge base to support semantic indexing, efficient retrieval, and grounded financial question answering.

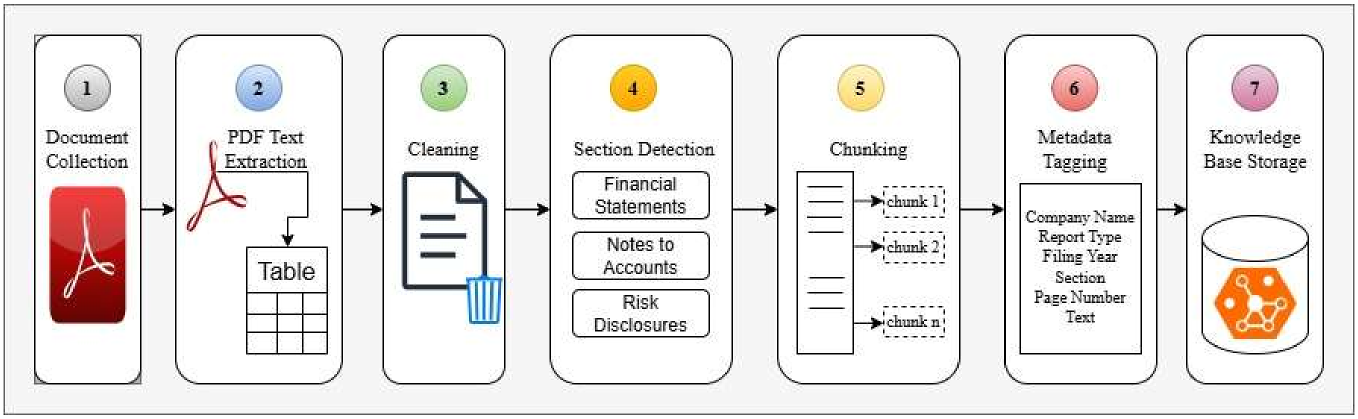


Figure 7.5.1 Financial Document Processing & Knowledge Base Preparation Workflow

Table 7.4 Financial Document Processing Workflow

Step	Description
PDF Text Extraction	Extract textual content from annual reports, quarterly filings, and financial PDFs
Table Extraction	Extract important financial tables and numerical statements where possible
Cleaning	Remove repeated headers, page numbers, formatting artifacts, and irrelevant text
Section Detection	Identify sections such as balance sheet, income statement, risk factors, and notes
Chunking	Split long documents into retrieval-ready semantic chunks
Metadata Tagging	Add company name, filing year, report type, page number, and section information
Storage	Store processed chunks and metadata within the retrieval knowledge base

Table 7.5 Sample Processed Financial Chunk

Field	Example
Chunk ID	AAPL_2022_AR_CH_015
Company	Apple
Year	2022
Report Type	Annual Report
Section	Consolidated Statements of Operations
Page Number	34
Text	Net sales increased to $394.3 billion in 2022, primarily driven by strong demand for iPhone, Mac, and Services across multiple geographic segments.

7.6 Agentic RAG Architecture Design and System Diagram

The proposed system is designed as an adaptive multi-agent retrieval and reasoning framework for financial document question answering. Instead of using one fixed workflow for every query, the architecture dynamically adjusts its retrieval and reasoning stages according to query complexity. This approach is intended to improve evidence retrieval, reasoning quality, and grounded answer generation when processing long financial reports such as annual filings, quarterly disclosures, and earnings documents.

The workflow begins with a Query Understanding Agent that identifies important contextual information such as company name, reporting period, financial terminology, and query complexity. Queries that appear ambiguous or retrieval-unfriendly are forwarded to a Query Refinement Agent for rewriting into clearer retrieval-oriented forms. A Retrieval Agent then searches the financial knowledge base using embeddings and metadata such as report type, filing year, section name, and page reference. Retrieved evidence is subsequently filtered to remove redundant or weakly relevant chunks before reasoning is performed across the selected financial context.

The filtered evidence is processed by a Reasoning Agent, where numerical interpretation, cross-section analysis, and contextual reasoning are performed before grounded responses are generated. The generated output is then validated through a Verification Agent to ensure consistency between retrieved evidence and final financial answers. The adaptive architecture allows simpler questions to follow shorter retrieval paths, while complex queries activate additional stages such as query refinement, multi-query expansion, evidence filtering, and deeper reasoning. This dynamic workflow is expected to improve retrieval precision, reasoning consistency, and answer reliability when compared with traditional fixed-pipeline systems.

Table 7.6 Agent Roles in Proposed System

Agent	Role in the Proposed System
Query Understanding Agent	Identifies query intent, company, financial terminology, reporting year, and complexity level
Query Refinement Agent	Rewrites broad or ambiguous financial questions into retrieval-friendly queries
Retrieval Agent	Retrieves relevant financial chunks and supporting evidence from the knowledge base
Evidence Filtering Agent	Removes weak, noisy, or redundant evidence before reasoning
Reasoning Agent	Performs financial reasoning and combines evidence from multiple sections
Answer Generation Agent	Produces the final grounded financial response
Verification Agent	Validates whether the answer is properly supported by retrieved evidence

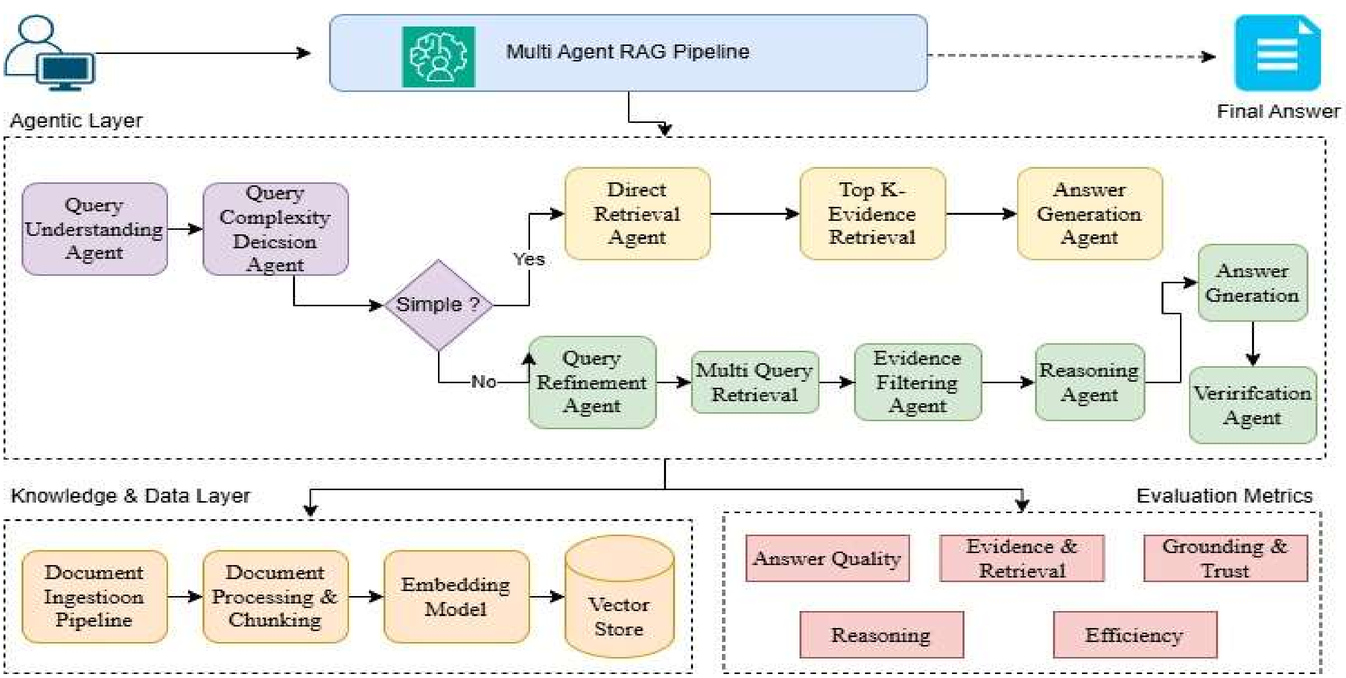


Figure 7.6.1 Agentic RAG Architecture

The proposed architecture follows an adaptive processing strategy rather than a fixed retrieval pipeline. Initially, the financial query is analysed to identify its intent, complexity level, company reference, reporting period, and important financial terminology. Based on this analysis, the system decides whether the query can be handled through a direct retrieval path or whether additional reasoning stages are required. Simpler questions that involve straightforward factual extraction are processed through a shorter workflow consisting mainly of retrieval and answer generation. In contrast, more complex questions activate additional stages such as query refinement, multi-query expansion, evidence filtering, and deeper financial reasoning. This adaptive routing mechanism allows the system to allocate computational effort according to query difficulty, improving retrieval precision and reasoning consistency while avoiding unnecessary processing for simpler financial questions.

7.7 Query Complexity Detection and Adaptive Routing

The proposed system includes an adaptive routing mechanism designed to classify financial questions according to their complexity before retrieval and reasoning are performed. The primary objective of this mechanism is to avoid processing every query through the same retrieval pipeline. Financial document question answering contains a wide range of query types. Certain questions involve direct factual extraction from a single section of a report, while others require comparison, aggregation, or reasoning across multiple financial statements and disclosures. A fixed retrieval workflow may therefore introduce unnecessary processing for simpler questions or insufficient reasoning for more complex analytical queries.

Within the proposed framework, financial questions are grouped into three complexity levels: simple, moderate, and complex. Simple queries generally involve direct numerical or factual lookup tasks where a single evidence chunk is sufficient to generate the answer. Moderate queries typically require comparison across reporting periods, interpretation of changes, or retrieval of multiple related values. Complex queries involve multi-step reasoning, trend interpretation, or explanation-oriented analysis where information may need to be collected from several sections of a financial report before a grounded response can be generated.

Table 7.7 Query Levels with Examples

Query Level	Meaning	Example
Simple	Direct factual or numerical lookup	What was Microsoft's revenue in 2022?
Moderate	Requires comparison or explanation	How did Apple's revenue change from 2021 to 2022?
Complex	Requires multi-step or multi-section reasoning	What factors contributed to the change in operating income?

The classification process is handled by the Query Understanding Agent. Multiple signals are analysed during this stage to estimate the reasoning difficulty and retrieval requirements of the incoming query. Company names, reporting years, financial metrics, and accounting terminology are identified to establish the financial context of the question. The query structure is also examined to determine whether the task involves factual lookup, comparison, trend analysis, explanation, or reasoning.

Table 7.8 Reasoning Signals

Signal	Meaning
Financial Entity	Identifies company name, reporting year, and financial metric
Question Type	Detects lookup, comparison, explanation, trend, or reasoning queries
Number of Required Values	Determines whether one or multiple values are needed
Need for Calculation	Checks whether arithmetic or comparison operations are required
Need for Multiple Evidence Chunks	Estimates whether information must be retrieved from different sections

After classification, the query is routed through an appropriate processing path. Simpler questions follow a shorter workflow that mainly involves direct retrieval and answer generation. More complex questions activate additional stages such as query refinement, evidence filtering, multi-query retrieval, and deeper financial reasoning. This adaptive routing strategy is intended to improve retrieval efficiency and reasoning consistency by matching the processing depth to the complexity of the financial query rather than applying a uniform pipeline to all questions.

Table 7.9 Adaptive Routing Decision Rules

Condition	Route
One company + one year + one metric	Simple route
Multiple years or comparison terms	Moderate route
Contains terms such as “why”, “explain”, “compare”, “trend”, or “reason”	Complex route
Requires values from different sections or statements	Complex route
Needs arithmetic calculation or financial interpretation	Moderate or complex route
Requires evidence from multiple pages or disclosures	Complex route

7.8 Retrieval, Evidence Selection, and Reasoning Workflow

After query classification and routing, retrieval and reasoning stages will be activated according to the complexity of the financial question. Simpler queries will use direct top-k semantic retrieval from the vector database, while more complex questions may involve query rewriting, query expansion, or multi-query retrieval to improve evidence coverage across different report sections. Retrieved chunks will then be evaluated by the Evidence Filtering Agent to remove redundant or weakly relevant information while preserving context that directly supports the query. Metadata such as company name, reporting year, page number, and section title may also be considered to improve contextual consistency and traceability. The filtered evidence will subsequently be forwarded to the Reasoning Agent for numerical interpretation, comparison, aggregation, and explanation-oriented reasoning.

The reasoning output will then be passed to the Answer Generation Agent to produce grounded natural language responses from the retrieved evidence. Before the final response is returned, the Verification Agent will perform an additional validation step to ensure consistency with the supporting financial evidence and reduce unsupported claims or numerical inconsistencies.

The overall retrieval and reasoning workflow followed within the proposed system is summarized below:

1. User financial question is received.
2. Query Understanding Agent identifies intent and query complexity.
3. Query Refinement Agent improves the question if refinement is required.
4. Retrieval Agent fetches relevant financial document chunks.
5. Evidence Filtering Agent selects the strongest supporting evidence.
6. Reasoning Agent combines evidence and performs financial reasoning.
7. Answer Generation Agent produces the final grounded response.
8. Verification Agent validates whether the generated answer is supported by retrieved evidence.

Table 7.10 Sample Evidence Selection Structure

Evidence ID	Company	Year	Section	Retrieved Text	Relevance Score
EV_001	Microsoft	2022	Consolidated Statements of Income	Revenue increased primarily due to continued growth in Intelligent Cloud and productivity services across enterprise customers	0.94
EV_002	Apple	2021	Management Discussion and Analysis	Net sales growth was driven by strong customer demand for iPhone products and expansion of service-based offerings.	0.91
EV_003	Tesla	2023	Risk Factors	Operating margins were negatively affected by supply chain disruptions, raw material pricing, and production cost fluctuations	0.88

The inclusion of evidence tracking within the workflow ensures that answers are generated based on retrieved financial context rather than unsupported language generation. This evidence-oriented design improves traceability, retrieval transparency, and reasoning reliability during financial document question answering.

7.9 Benchmarking Framework and Baseline Comparison

The proposed adaptive multi-agent retrieval and reasoning framework will be evaluated through a controlled benchmarking strategy against multiple baseline systems. The primary objective of this comparison is to determine whether adaptive routing, query refinement, evidence filtering, and multi-agent reasoning improve financial document question answering performance when compared with simpler retrieval pipelines. Rather than evaluating the proposed system in isolation, the study is designed to measure how each additional retrieval and reasoning component contributes to overall performance improvement.

The benchmarking process begins with a direct language model baseline where answers are generated without any retrieval stage. This baseline helps measure whether external financial document retrieval is necessary for accurate question answering. A second baseline uses a simple retrieval pipeline consisting of direct top-k retrieval followed by answer generation. Additional baselines progressively introduce more advanced retrieval strategies such as query rewriting and multi-query retrieval. The final system incorporates adaptive routing, multiple specialized agents, evidence filtering, reasoning, and answer verification.

Table 7.11 Benchmarking Structure

System	Description	Purpose
Baseline 1: Direct LLM	Language model answers without retrieval	Evaluates whether retrieval is necessary
Baseline 2: Naïve RAG	Simple top-k retrieval followed by answer generation	Provides basic retrieval baseline
Baseline 3: Query-Rewritten RAG	Query rewriting applied before retrieval	Evaluates impact of query refinement
Baseline 4: Multi-Query RAG	Multiple query variations generated before retrieval	Tests improvement in retrieval coverage
Proposed System: Adaptive Multi-Agent RAG	Adaptive routing with specialized retrieval, reasoning, filtering, and verification agents	Evaluates the complete proposed framework

To ensure fairness and experimental consistency, all systems will be evaluated under the same controlled environment. Only the retrieval and reasoning pipeline design will vary between experiments, while all other supporting components remain fixed. This controlled comparison helps ensure that performance differences can be attributed primarily to architectural changes rather than variations in datasets, embeddings, or retrieval infrastructure.

Table 7.12 Controlled Experimental Setup

Controlled Component	Fixed Setup
Dataset	FinanceBench
Financial Documents	Same PDF collection
Language Model	Same base model
Embedding Model	Same embedding configuration
Vector Database	Same vector storage system
Chunk Size	Same chunking configuration
Top-k Retrieval Value	Same retrieval limit where applicable
Benchmark Questions	Same evaluation question set
Evaluation Metrics	Same evaluation methodology and scoring metrics

The benchmarking structure is intentionally progressive in nature. Each stage introduces one additional improvement over the previous system, allowing the contribution of individual components to be analysed more systematically. The evaluation flow therefore follows the sequence:

Direct Language Model → Naïve Retrieval Pipeline → Query Rewriting → Multi-Query Retrieval → Adaptive Multi-Agent Retrieval and Reasoning

The progression strengthens the overall research design since it demonstrates not only whether the final system performs better, but also why specific retrieval and reasoning enhancements contribute to improved financial question answering performance.

7.10 Evaluation Metrics and Performance Measurement

The proposed system will be evaluated using multiple groups of metrics that measure not only answer correctness but also retrieval quality, evidence grounding, reasoning consistency, and system efficiency. In financial document question answering, generating a correct-looking response is not sufficient if the answer is not properly supported by financial evidence. Financial analysis tasks often involve numerical interpretation, multi-section reasoning, and evidence-sensitive decision making. For this reason, the evaluation framework is designed to assess the complete retrieval and reasoning workflow rather than focusing only on final answer generation.

A. Answer Quality Metrics

These metrics evaluate whether the generated response correctly addresses the financial question and matches the expected answer content.

Table 7.13 Answer Quality Metrics

Metric	Purpose	
Answer Relevance	Measures whether the generated answer directly addresses the financial question	$\text{Answer Relevance}=\frac{Q \cdot A}{\lvert Q \rvert \lvert A \rvert}$
Exact Match	Evaluates whether the generated answer exactly matches the reference answer, especially for numerical queries	$EM=\begin{cases}1 & \text{Generated Answer = Reference Answer}\\ 0 & \text{otherwise}\end{cases}$
F1-Score	Measures token-level overlap between generated and reference answers	$F1=\frac{2\times \text{Precision}\times \text{Recall}}{\text{Precision} + \text{Recall}}$
Semantic Similarity	Evaluates meaning-level similarity between generated and expected responses	$\text{Semantic Similarity}=\frac{E_g \cdot E_r}{\lvert E_g \rvert \lvert E_r \rvert}$

B. Evidence and Retrieval Metrics

These metrics evaluate whether the retrieval process successfully identifies relevant supporting evidence from financial documents.

Table 7.14 Evidence & Retrieval Metrics

Metric	Purpose	
Context Recall	Measures whether the required supporting evidence was successfully retrieved	$\text{Context Recall}=\frac{\text{Relevant Chunks Retrieved}}{\text{Total Relevant Chunks}}$
Context Precision	Evaluates whether retrieved chunks are relevant to the financial query	$\text{Context Precision}=\frac{\text{Relevant Retrieved Chunks}}{\text{Total Retrieved Chunks}}$
Hit@K	Checks whether the correct evidence appears within the top-k retrieved chunks	$\text{Hit @K}=\frac{\text{Queries With Correct Evidence In Top K}}{\text{Total Queries}}$ K = number of top retrieved chunks considered
Mean Reciprocal Rank (MRR)	Measures the ranking position of the first correct evidence chunk	$MRR=\frac{1}{N}\sum_{i=1}^{N}\frac{1}{\text{rank}_i}$

C. Grounding and Trust Metrics

These metrics focus on evidence support, factual grounding, and reduction of unsupported financial claims.

Table 7.15 Grounding & Trust Metrics

Metric	Purpose	
Faithfulness	Measures whether the generated answer is fully supported by retrieved evidence	$\text{Faithfulness} = \frac{\text{Supported Statements}}{\text{Total Statements In Generated Answer}}$
Hallucination Rate	Evaluates the frequency of unsupported or incorrect financial claims	$\text{Hallucination Rate} = \frac{\text{Unsupported Claims}}{\text{Total Claims}}$
Evidence Coverage	Checks whether all important evidence points are incorporated into the final response	$\text{Evidence Coverage} = \frac{\text{Evidence Points Used}}{\text{Total Relevant Evidence Points}}$
Citation Correctness	Verifies whether cited evidence actually supports the generated answer	$\text{Citation Correctness} = \frac{\text{Correct Supporting Citations}}{\text{Total Citations}}$

D. Financial Reasoning Metrics

These metrics evaluate whether the system performs correct financial reasoning and numerical interpretation.

Table 7.16 Financial Reasoning Metrics

Metric	Purpose	
Numerical Accuracy	Measures correctness of extracted financial values	$\text{Numerical Accuracy} = \frac{\text{Correct Numerical Values}}{\text{Total Numerical Values}}$
Calculation Accuracy	Evaluates arithmetic operations, comparisons, and derived calculations	$\text{Calculation Accuracy} = \frac{\text{Correct Calculations}}{\text{Total Calculations}}$
Multi-step Reasoning Score	Measures whether multiple evidence points are combined correctly	$\text{Multi Step Reasoning} = \frac{\text{Correct Reasoning Steps}}{\text{Total Reasoning Steps}}$
Explanation Completeness	Evaluates whether the reasoning process is explained clearly and consistently	$\text{Explanation Completeness} = \frac{\text{Covered Reasoning Points}}{\text{Total Expected Reasoning Points}}$

E. Efficiency Metrics

These metrics evaluate computational efficiency and practical usability of the proposed system.

Table 7.17 Efficiency Metrics

Metric	Purpose	
Latency	Measures total response generation time	$\text{Latency} = \text{Response End Time} - \text{Response Start Time}$
Token Usage	Measures input and output token consumption	$\text{Token Usage} = \text{Input Tokens} + \text{Output Tokens}$
Retrieval Time	Measures evidence search and retrieval duration	$\text{Retrieval Time} = \text{Retrieval End Time} - \text{Retrieval Start Time}$
Cost per Answer	Estimates computational or API cost associated with generating each response	$\text{Cost Per Answer} = \frac{\text{Total API Or Computation Cost}}{\text{Total Generated Answers}}$

This broader evaluation strategy is expected to provide a more realistic assessment of financial document question answering performance, particularly when comparing adaptive multi-agent systems with traditional fixed retrieval pipelines.

7.11 End-to-End Implementation Workflow

The proposed study follows a structured end-to-end implementation workflow beginning from financial document collection and continuing through retrieval, reasoning, benchmarking, and evaluation. The workflow is designed to ensure that all baseline systems and the proposed adaptive multi-agent framework are evaluated under consistent experimental conditions. Since financial document question answering involves both retrieval quality and reasoning reliability, the implementation process is organized to support document grounding, evidence tracking, and controlled benchmarking across multiple retrieval strategies.

Step 1 - Collect FinanceBench documents

Download financial PDFs and question-answer evidence files from the FinanceBench dataset.

Step 2 - Extract text and tables from financial PDFs

Extract report text, tables, sections, page numbers, and financial statements.

Step 3 - Clean and structure the extracted content

Remove repeated headers, page numbers, noisy symbols, and irrelevant text.

Step 4 - Create financial document chunks

Split reports into meaningful chunks using section-aware chunking.

Step 5 - Add metadata to chunks

Store company name, year, report type, section, page number, and chunk ID.

Step 6 - Generate embeddings and store in vector DB

Convert chunks into embeddings and store them in a vector database.

Step 7 -Prepare benchmark questions

Use FinanceBench questions and classify them into simple, moderate, and complex levels.

Step 8 - Build baseline systems

Implement Direct LLM, Naïve RAG, Query-Rewritten RAG, and Multi-Query RAG.

Step 9 - Build proposed adaptive multi-agent RAG system

Implement query understanding, query refinement, retrieval, reasoning, generation, and verification agents.

Step 10 - Run all benchmark questions

Pass the same questions to all systems.

Step 11 - Store generated answers and retrieved evidence

Save output answer, retrieved chunks, evidence IDs, model name, system type, and latency.

Step 12 - Evaluate results using metrics

Calculate answer relevance, faithfulness, context recall, context precision, numerical accuracy, and latency.

Step 13 - Compare baseline and proposed system

Analyse whether adaptive multi-agent RAG performs better than fixed RAG pipelines.

Step 14 - Perform error analysis

Review failed cases such as wrong numbers, missing evidence, weak reasoning, or hallucinated claims.

Step 15 - Summarise best-performing approach

Identify which system gives the best balance of accuracy, grounding, reasoning, and efficiency.

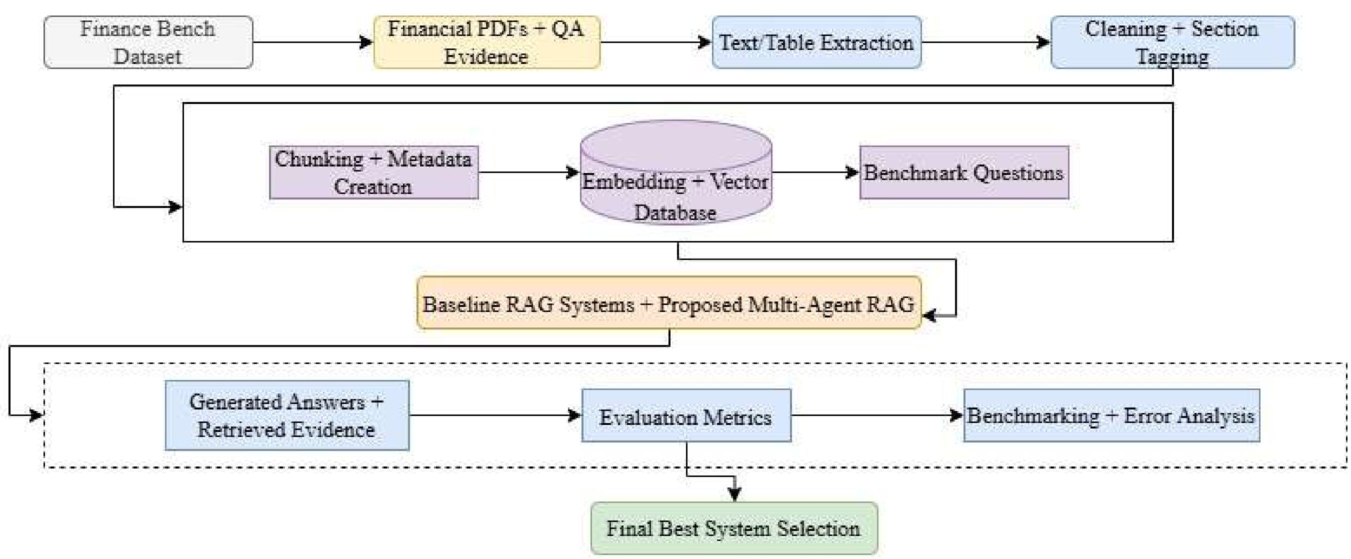


Figure 7.11.1 End-to-End Implementation Workflow

7.12 Methodology Summary

The methodology of adaptive multi-agent financial document question answering system using the FinanceBench dataset contains workflow of document preprocessing, chunk generation, semantic retrieval, adaptive query routing, evidence selection, reasoning, answer generation, and verification stages. Multiple baseline RAG pipelines will be evaluated under identical conditions using metrics such as answer relevance, faithfulness, numerical correctness, reasoning consistency, and retrieval effectiveness.

8. Requirements Resources

8.1 Hardware Requirements

- Intel Core i7 or AMD Ryzen 7 processor for smooth system performance.
- Minimum 16 GB RAM for document processing and retrieval tasks.
- NVIDIA GPU such as RTX 3060 or higher for faster model execution.
- At least 500 GB SSD storage for datasets, embeddings, and outputs.
- Stable internet connection for downloading datasets and API access.
- Windows 11 or Ubuntu Linux operating system for framework compatibility.

8.2 Software Requirements

Table 8.1 Software Requirements

Software	Version	Purpose
Python	3.12	Core programming and system development
Visual Studio Code	1.99 or later	Code development and debugging
Jupyter Notebook	7.4 or later	Experiment execution and analysis
LangChain	0.3 or later	Agent workflow and retrieval pipeline development
ChromaDB	0.5 or later	Vector database storage and retrieval
Pandas	2.2 or later	Data preprocessing and analysis
NumPy	2.2 or later	Numerical computation support
Scikit-learn	1.6 or later	Evaluation and similarity calculations
FAISS	1.10 or later	Fast similarity search and vector indexing

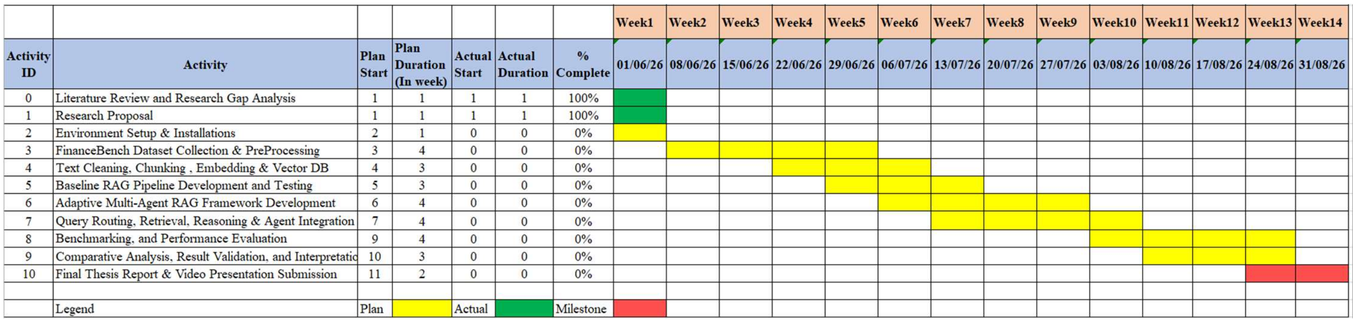
8.3 Dataset Requirements

- FinanceBench dataset containing financial question-answer pairs with supporting evidence from real company reports.
- Financial PDF documents such as annual reports, quarterly filings, and financial disclosures for retrieval and reasoning experiments.
- Benchmark questions categorized into simple, moderate, and complex financial queries for controlled evaluation.

9. Research Plan

9.1 Gantt Chart

The Gantt chart outlines the weekly research schedule from June to August 2026, covering dataset preparation, preprocessing, RAG framework development, experimentation, evaluation, and thesis documentation. It also illustrates overlapping research activities and the planned timeline for analysis, validation, review, and final submission.



Activity ID	Activity	Plan Start	Plan Duration (In week)	Actual Start	Actual Duration	% Complete	Week1 01/06/26	Week2 08/06/26	Week3 15/06/26	Week4 22/06/26	Week5 29/06/26	Week6 06/07/26	Week7 13/07/26	Week8 20/07/26	Week9 27/07/26	Week10 03/08/26	Week11 10/08/26	Week12 17/08/26	Week13 24/08/26	Week14 31/08/26
0	Literature Review and Research Gap Analysis	1	1	1	1	100%														
1	Research Proposal	1	1	1	1	100%														
2	Environment Setup & Installations	2	1	0	0	0%														
3	FinanceBench Dataset Collection & PreProcessing	3	4	0	0	0%														
4	Text Cleaning, Chunking , Embedding & Vector DB	4	3	0	0	0%														
5	Baseline RAG Pipeline Development and Testing	5	3	0	0	0%														
6	Adaptive Multi-Agent RAG Framework Development	6	4	0	0	0%														
7	Query Routing, Retrieval, Reasoning & Agent Integration	7	4	0	0	0%														
8	Benchmarking, and Performance Evaluation	9	4	0	0	0%														
9	Comparative Analysis, Result Validation, and Interpretatic	10	3	0	0	0%														
10	Final Thesis Report & Video Presentation Submission	11	2	0	0	0%														
	Legend	Plan		Actual		Milestone														

Figure 9.1.1 Gantt Chart

9.2 Risk and Contingency Plans

Table 9.1 Risk and Contingency Plans

Risk	Contingency Plan
Large financial PDFs increasing processing and retrieval time	Document chunking, metadata indexing, and optimized preprocessing will be used to improve retrieval efficiency.
Limited computational resources during embedding generation or inference	Lightweight embedding models and cloud platforms such as Google Colab will be utilized when required.
Retrieval of irrelevant evidence or hallucinated responses	Evidence filtering, metadata-aware retrieval, and verification stages will be applied to improve answer grounding.
Delays during experimentation or framework integration	Intermediate checkpoints, automated backups, and buffer weeks will be maintained to manage unexpected delays.

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